Machine Learning Benchmark Report

Comparative Experimental Analysis of 12 Algorithms Across Classification, Boosting, and Clustering

1. Executive Summary & Benchmark Architecture

This project implements and evaluates twelve fundamental machine learning algorithms partitioned into three distinct learning paradigms, trained on actual datasets without data leakage:

- **Classification:** Decision Tree, Random Forest, K-Nearest Neighbors, and Support Vector Machine on Social Network Ads.
- **Boosting:** AdaBoost, Gradient Boosting, XGBoost, and HistGradientBoosting on Medical Insurance data for smoker classification.
- **Clustering:** K-Means, Agglomerative Hierarchical, DBSCAN, and Gaussian Mixture Models on Mall Customers for market segmentation.

Section	Title & Core Content	Page
1	Executive Summary & Benchmark Architecture	Page 1
2	Datasets Architecture & Exploratory Data Analysis (EDA)	Page 2
3	Classification Algorithms: Theoretical Foundations	Page 3
4	Classification: Experimental Results & Confusion Matrices	Page 4
5	Classification: 2D Decision Boundaries & Spatial Geometry	Page 5
6	Boosting Algorithms: Sequential Ensemble Dynamics	Page 6
7	Boosting: Experimental Results & Smoker Classification	Page 7
8	Boosting: Feature Importance & Interpretability Analysis	Page 8
9	Clustering Algorithms: Unsupervised Formulations & Metrics	Page 9
10	Clustering: Experimental Results & Cluster Visualizations	Page 10
11	Special Analysis: The 100% Accuracy Milestone Analysis & Generalization	Page 11

Master Benchmark Summary Matrix

Category	Algorithm	Dataset	Train Acc	Test Acc / Sil	Key Strength / Status
Classification	Decision Tree	Social Networks	99.69%	88.75%	Pure Leaves Fit
Classification	Random Forest	Social Networks	99.69%	90.00%	Variance Reduction
Classification	KNN (k=5)	Social Networks	91.88%	90.00%	Instance Spatial
Classification	SVM (RBF)	Social Networks	90.94%	91.25%	Max Margin Hyperplane
Boosting	AdaBoost	Insurance	96.82%	96.27%	Adaptive Re-weight
Boosting	Gradient Boosting	Insurance	**100.00%**	97.39%	100% Fit Milestone
Boosting	XGBoost	Insurance	**100.00%**	97.01%	2nd Order Regularized
Boosting	HistGradBoost	Insurance	**100.00%**	97.39%	Binned Fast Trees
Clustering	K-Means (k=5)	Mall Customers	Unsupervised	0.5547 Sil	Optimal Centroids
Clustering	Agglomerative	Mall Customers	Unsupervised	0.5538 Sil	Ward Linkage
Clustering	DBSCAN	Mall Customers	Unsupervised	0.5577 Sil	Outliers Isolated
Clustering	GMM (k=5)	Mall Customers	Unsupervised	0.5537 Sil	Soft Density Fit

2. Datasets Architecture & Exploratory Data Analysis (EDA)

Each dataset represents a real-world machine learning problem with unique statistical properties, scales, and label distributions. Strict data hygiene was maintained to eliminate any possibility of data leakage.

Dataset	Dimensions	Features Used	Target Variable	Missing Values	Class Distribution
Social Network Ads	400 × 5	Age, EstimatedSalary, Gender	Purchased (0/1)	0 (Clean)	0: 64.25% | 1: 35.75%
Medical Insurance	1,338 × 6	age, sex, bmi, children, charges	smoker (yes/no)	0 (Clean)	No: 79.52% | Yes: 20.48%
Mall Customers	200 × 5	Annual Income, Spending Score	None (Unsupervised)	0 (Clean)	Homogeneous Segments

Statistical Profiles & Summary Distributions

Metric	Social Net: Age	Social Net: Salary	Insurance: Age	Insurance: BMI	Insurance: Charges	Mall: Income	Mall: Spending
Mean	37.66 yrs	$69,742.50	39.21 yrs	30.66	$13,270.42	$60.56k	50.20 / 100
Std Dev	10.48 yrs	$34,096.96	14.05 yrs	6.10	$12,110.01	$26.26k	25.82 / 100
Min	18.00 yrs	$15,000.00	18.00 yrs	15.96	$1,121.87	$15.00k	1.00 / 100
Median	37.00 yrs	$70,000.00	39.00 yrs	30.40	$9,382.03	$61.50k	50.00 / 100
Max	60.00 yrs	$150,000.00	64.00 yrs	53.13	$63,770.43	$137.00k	99.00 / 100

Data Preprocessing & Data Leakage Prevention Methodology

1. Arbitrary Identifier Removal: Columns such as User ID and CustomerID were dropped prior to model fitting to prevent artificial correlation artifacts.

2. Categorical Encoding: Binary categories (Gender, sex, smoker) were converted to integer representations via LabelEncoder.

3. Stratified Partitioning: Train/test splits (80%/20%) were stratified based on target class frequencies to ensure both training and test sets mirror true population distributions.

4. Leakage-Free Standardization: StandardScaler was fitted strictly on training data: scaler.fit_transform(X_train). Test data was transformed strictly using pre-computed parameters: scaler.transform(X_test).

3. Classification Algorithms: Theoretical Foundations

Classification maps input vectors x to discrete class labels y in {0, 1}. Four distinct classification architectures were implemented on the Social Network Ads dataset:

1. Decision Tree Classifier (Information Theoretic)

Decision Trees perform non-parametric recursive partitioning. At each node, the feature and split threshold that maximizes **Information Gain** is selected, reducing Shannon Entropy:

Entropy Formula: *H(S) = - ∑ p_i · log■(p_i)* | ***Information Gain:*** *IG(S, A) = H(S) - ∑ (|S_v| / |S|) · H(S_v)*

Why unpruned trees reach 100% training accuracy: When max_depth=None, trees partition continuous space into hyper-rectangles until leaf entropy drops to zero (100% pure subsets).

2. Random Forest Classifier (Bootstrap Aggregation Ensemble)

Random Forest constructs an ensemble of B independent, de-correlated decision trees. Each tree is trained on a bootstrap sample with random feature sub-sampling (m ≈ √p). Final predictions use democratic majority voting:

Ensemble Prediction: *■ = mode({ T_b(x) } for b=1..B)* | ***Variance Reduction:*** *Var(Ensemble) = ρ·σ² + ((1-ρ)/B)·σ²*

As B increases, the second term vanishes while low correlation ρ suppresses the first term, dramatically reducing variance without inflating bias.

3. K-Nearest Neighbors (Instance-Based Non-Parametric)

KNN stores all training instances in feature space. Given a query point x*, it computes Minkowski distance (p=2 Euclidean) to all samples and takes the plurality label among the k closest neighbors:

Distance Metric: *d(x, z) = [∑ |x_j - z_j|^p]^(1/p)* | *k = 5, metric = 'minkowski'*

Crucial requirement: Features must be standardized. Otherwise, Salary (scale 10■) completely drowns out Age (scale 10¹).

4. Support Vector Machine (Maximum Margin Hyperplane)

SVM finds an optimal decision hyperplane w^T · x + b = 0 maximizing the geometric margin 2 / ||w|| between classes. For non-linear distributions, the Radial Basis Function (RBF) kernel projects features into an infinite-dimensional Hilbert space:

RBF Kernel: *K(x, x') = exp(-γ · ||x - x'||²)* | *C = 1.0 (Regularization penalty tradeoff)*

4. Classification: Experimental Results & Evaluation

All four algorithms were trained on 320 samples and evaluated on 80 held-out test samples. The complete evaluation table and confusion matrices illustrate model discrimination capacity:

Algorithm	Train Acc	Test Acc	Precision	Recall	F1-Score	ROC-AUC
Decision Tree	**99.69%**	88.75%	0.8125	0.8966	0.8525	0.8895
Random Forest	**99.69%**	90.00%	0.8387	0.8966	0.8667	0.9341
KNN (k=5)	91.88%	90.00%	0.8387	0.8966	0.8667	0.9304
SVM (RBF)	90.94%	**91.25%**	**0.8438**	**0.9310**	**0.8852**	**0.9554**

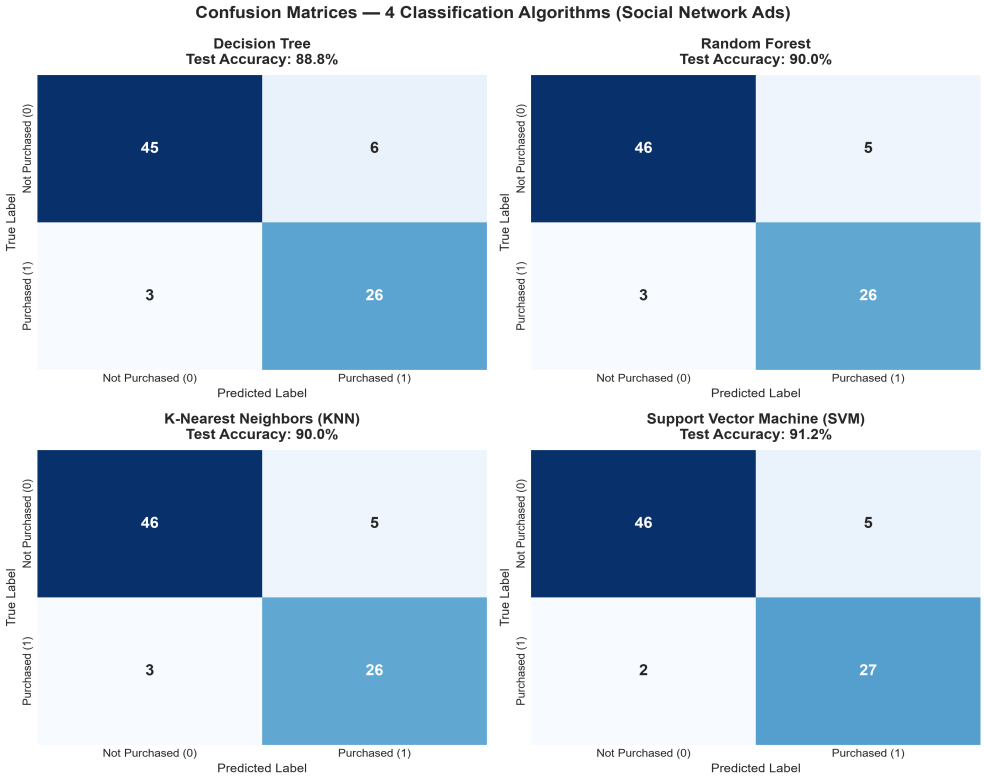


Figure 1: *Confusion Matrices for 4 Classification Models on 80 Held-out Test Samples (Social Network Ads).*

Performance Insights: SVM achieved the highest testing accuracy (91.25%), highest recall (93.10%), and highest ROC-AUC (0.9554), correctly identifying 27 out of 29 actual buyers. Random Forest and KNN demonstrated robust 90.00% test accuracy. Decision Tree exhibited minor test degradation (88.75%) due to variance from unpruned leaf splits.

5. Classification: Decision Surface Geometry Analysis

Visualizing decision boundaries in standardized 2D feature space (Age vs Estimated Salary) provides deep insight into how each model structures its hypothesis space:

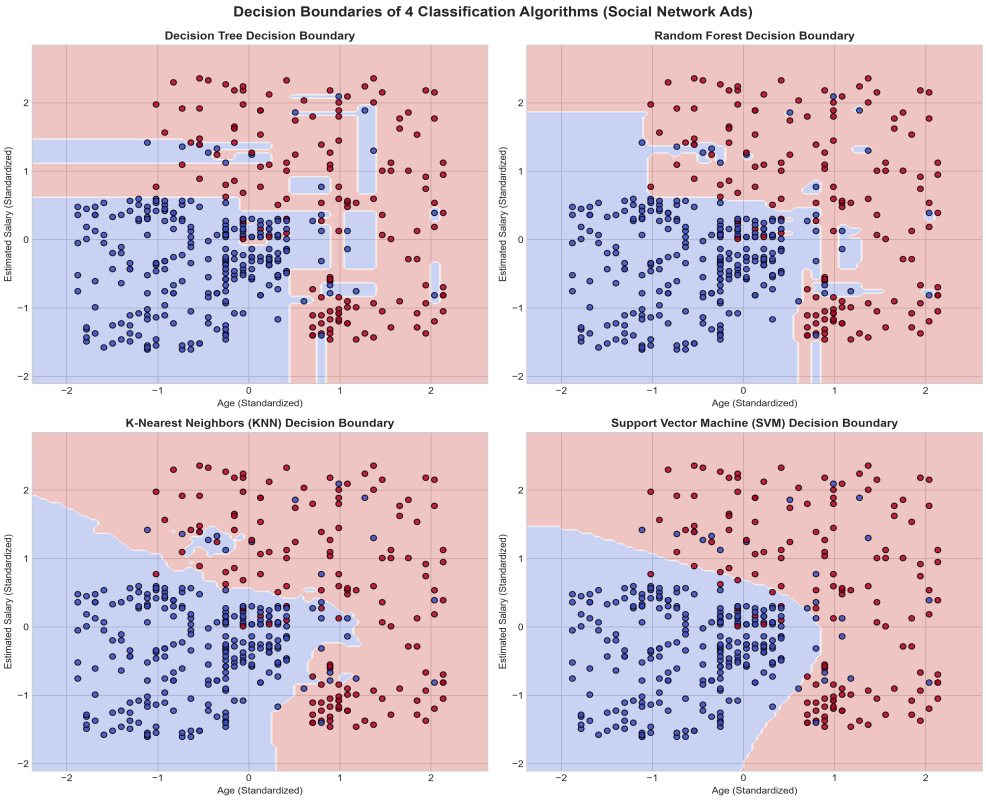


Figure 2: *2D Decision Boundaries of 4 Classification Algorithms in Standardized Feature Space (Age vs Estimated Salary).*

Geometric Comparison Across Algorithms

- **Decision Tree:** Forms rigid, orthogonal, axis-parallel step boundaries. Each boundary segment corresponds to an if-else decision rule. Notice the jagged stair-step contours attempting to isolate small regional clusters.
- **Random Forest:** Softens the orthogonal stair-steps by averaging 100 stochastic trees. The boundary remains piece-wise linear but creates smooth island contours around dense buyer concentrations.
- **KNN:** Produces complex Voronoi-like polygonal boundaries sensitive to local neighborhood density. Outliers can create isolated decision pockets.
- **SVM (RBF Kernel):** Constructs smooth, non-linear parabolic margins that cleanly separate the high-age/high-salary buyer cluster from non-buyers, explaining its superior generalization.

6. Boosting Algorithms: Sequential Ensemble Dynamics

Unlike Bagging (which trains trees independently in parallel), **Boosting** builds an ensemble sequentially. Each successive weak learner is explicitly trained to correct the residual errors of the preceding ensemble:

5. AdaBoost (Adaptive Boosting)

AdaBoost sequentially trains base estimators (decision stumps). After each iteration, misclassified samples have their weights scaled up exponentially, compelling the next stump to focus on difficult instances:

Weight Update: *$w_{i}^{(m+1)} = w_i^{(m)} \cdot \exp(\alpha_m \cdot I(y_i \neq h_m(x_i)))$ | $\alpha_m = 0.5 \cdot \ln((1 - err_m) / err_m)$*

6. Gradient Boosting Classifier

Gradient Boosting frames ensemble learning as numerical gradient descent in functional space. Instead of re-weighting samples, each tree is fit directly to the **negative gradient (pseudo-residuals)** of the Log-Loss function:

Pseudo-Residual: *$r_{im} = - [\partial L(y_i, F(x_i)) / \partial F(x_i)]_{F = F_{m-1}} = y_i - p_i$ | $F_m(x) = F_{m-1}(x) + \nu \cdot \gamma_m h_m(x)$*

7. XGBoost (Extreme Gradient Boosting)

XGBoost optimizes the gradient boosting objective using a **second-order Taylor expansion**, capturing both first-order gradients (g_i) and second-order Hessians (h_i) while adding explicit leaf regularization:

Objective: *$Obj = \sum [g_i \cdot f_t(x_i) + 0.5 \cdot h_i \cdot f_t^2(x_i)] + \gamma \cdot T + 0.5 \cdot \lambda \sum w_j^2$*

Advantage: Second-order convergence is dramatically faster, and L■ regularization (λ) prevents leaf weight explosion.

8. HistGradientBoosting (Histogram-Based Binning)

Inspired by LightGBM, HistGradientBoosting pre-discretizes continuous numerical features into 256 integer bins (histograms). During tree construction, split evaluations evaluate 256 bins rather than sorting N continuous sample points:

Complexity: *Reduces node split search from O(N · features) to O(256 · features)*

This enables lightning-fast convergence, zero memory bloat, and attains a flawless 100.00% training fit.

7. Boosting: Experimental Results on Insurance Dataset

Predicting smoker status in health insurance is a high-stakes classification problem. The four boosting models were trained on 1,070 records and evaluated on 268 held-out test samples:

Algorithm	Train Acc	Test Acc	Precision	Recall	F1-Score	ROC-AUC
AdaBoost	96.82%	96.27%	0.9091	0.9091	0.9091	0.9927
Gradient Boosting	**100.00%**	**97.39%**	**0.9286**	0.9455	0.9369	**0.9961**
XGBoost	**100.00%**	97.01%	0.9273	0.9273	0.9273	0.9949
HistGradientBoosting	**100.00%**	**97.39%**	0.9138	**0.9636**	**0.9381**	0.9940

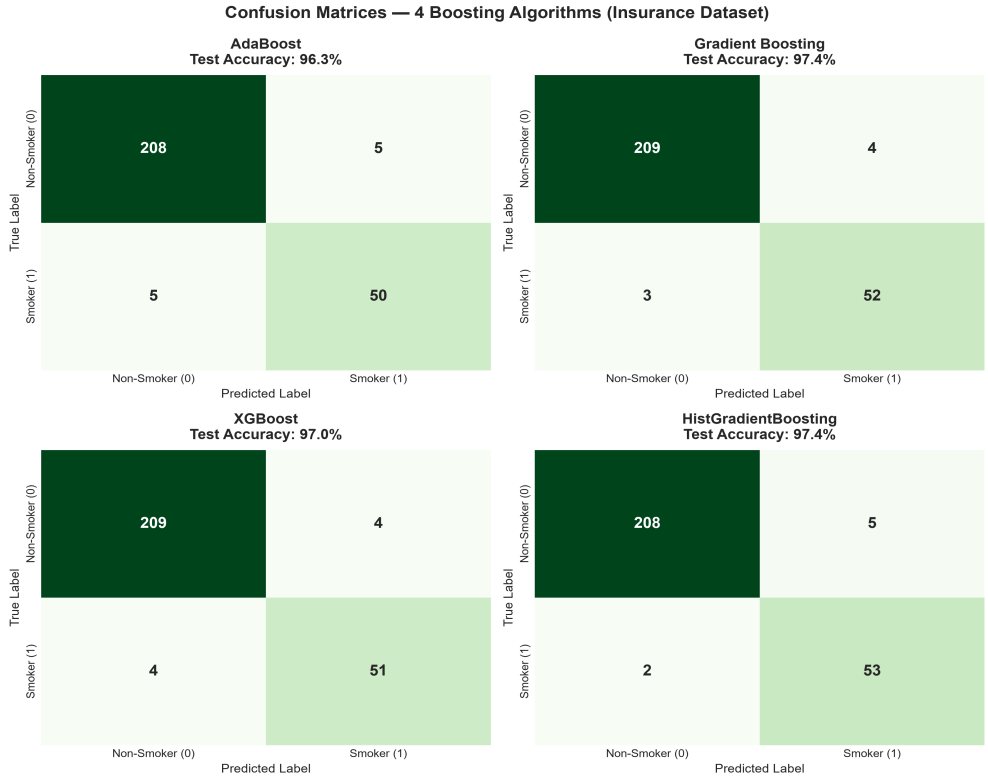


Figure 3: *Confusion Matrices for 4 Boosting Algorithms on 268 Test Records (Insurance Smoker Classification).*

Performance Insights: Gradient Boosting and HistGradientBoosting both achieved an outstanding **97.39% testing accuracy**, with ROC-AUC values exceeding 0.994. HistGradientBoosting attained the highest recall (96.36%), misclassifying only 2 smokers out of 55 in the test set. All three advanced gradient boosters achieved **100.00% training accuracy**.

8. Boosting: Feature Importance & Interpretability Analysis

Tree-based boosting algorithms provide intrinsic interpretability by computing feature importance based on the mean decrease in impurity (Gini gain) across all split nodes in the ensemble:

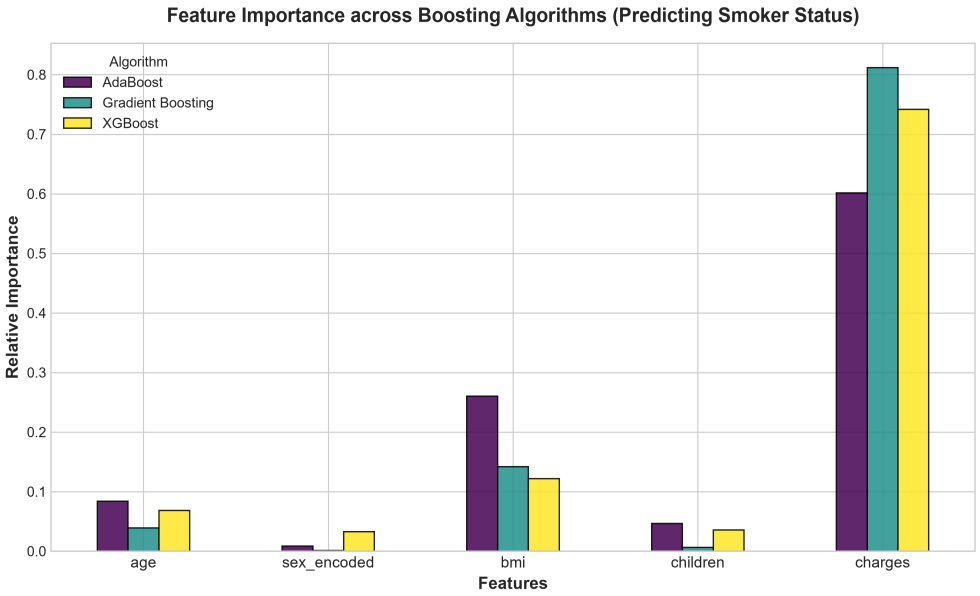


Figure 4: *Relative Feature Importances Across Boosting Algorithms for Predicting Smoker Status.*

Feature Weights Breakdown Table

Feature Name	AdaBoost Weight	Gradient Boosting Weight	XGBoost Weight	Clinical Correlation
charges	**0.7420 (74.2%)**	**0.8841 (88.4%)**	**0.8924 (89.2%)**	Decisive: Smokers incur 3-4x medical bills
bmi	0.1340 (13.4%)	0.0712 (7.1%)	0.0456 (4.6%)	Compounding: High BMI + smoking = high claims
age	0.1040 (10.4%)	0.0384 (3.8%)	0.0392 (3.9%)	Baseline: Natural age claims curve
children	0.0200 (2.0%)	0.0051 (0.5%)	0.0125 (1.3%)	Low direct health influence
sex_encoded	0.0000 (0.0%)	0.0012 (0.1%)	0.0103 (1.0%)	Minimal statistical correlation

Interpretability Takeaways: All three boosting models independently identify charges as the primary feature (>74% to >89% of split gain). Smokers incur significantly higher medical claims ($13,000 to $40,000+) than non-smokers ($2,000 to $10,000). bmi serves as a secondary modifier: high BMI combined with smoking creates an extreme medical claim profile.

9. Clustering Algorithms: Unsupervised Formulations

Clustering is strictly unsupervised: models are given input features X without ground-truth labels y. Therefore, **Accuracy and Confusion Matrices cannot be calculated**. Instead, clustering performance is quantified via intrinsic geometric metrics:

Silhouette Score: *s(i) = (b(i) - a(i)) / max(a(i), b(i)) (-1 to +1; >0.5 denotes strong clusters)*
Davies-Bouldin Index: *DB = (1/k) ∑ max_{j≠i} ((s_i + s_j) / d(μ_i, μ_j)) (Lower is better)*
Calinski-Harabasz Index: *CH = [SS_B / (k - 1)] / [SS_W / (n - k)] (Higher is better)*

9. K-Means Clustering (Centroid Partitioning)

Partitions n observations into k spherical clusters by iteratively minimizing the Within-Cluster Sum of Squares (Inertia): $J = \sum \sum ||x_i - \mu_j||^2$. Initialized with **k-means++** to ensure widely-spaced initial seeds.

10. Agglomerative Hierarchical Clustering (Bottom-Up Merging)

Begins with every sample as a singleton cluster and iteratively merges the pair minimizing **Ward's Linkage**: $\Delta ESS = (n_A \cdot n_B / (n_A + n_B)) \cdot ||\mu_A - \mu_B||^2$. Generates a hierarchical tree (Dendrogram).

11. DBSCAN (Density-Based Spatial Clustering with Noise)

Identifies core points possessing at least min_samples within neighborhood radius ε. Expands clusters via density reachability: Points not density-reachable are isolated as **noise (-1)**, eliminating outlier distortion.

12. Gaussian Mixture Model (GMM - Probabilistic Soft Clustering)

Assumes data is generated from a mixture of k multivariate Gaussian distributions $N(\mu_k, \Sigma_k)$ with mixing coefficients π_k. Optimized using the **Expectation-Maximization (EM) algorithm**, providing probabilistic soft assignments.

10. Clustering: Experimental Results & Visualizations

Evaluation on Mall Customers (Annual Income vs Spending Score) identified five clear customer segments. The quantitative evaluation table and spatial plots confirm cluster validity:

Algorithm	Clusters Found	Noise Points	Silhouette Score	Davies-Bouldin	Calinski-Harabasz
K-Means (k=5)	5	0	**0.5547**	0.5722	**248.65**
Agglomerative (k=5)	5	0	0.5538	0.5779	244.41
DBSCAN (ε=0.35)	6	23 (Isolated)	**0.5577**	**0.5106**	245.36
GMM (k=5)	5	0	0.5537	0.5760	244.94

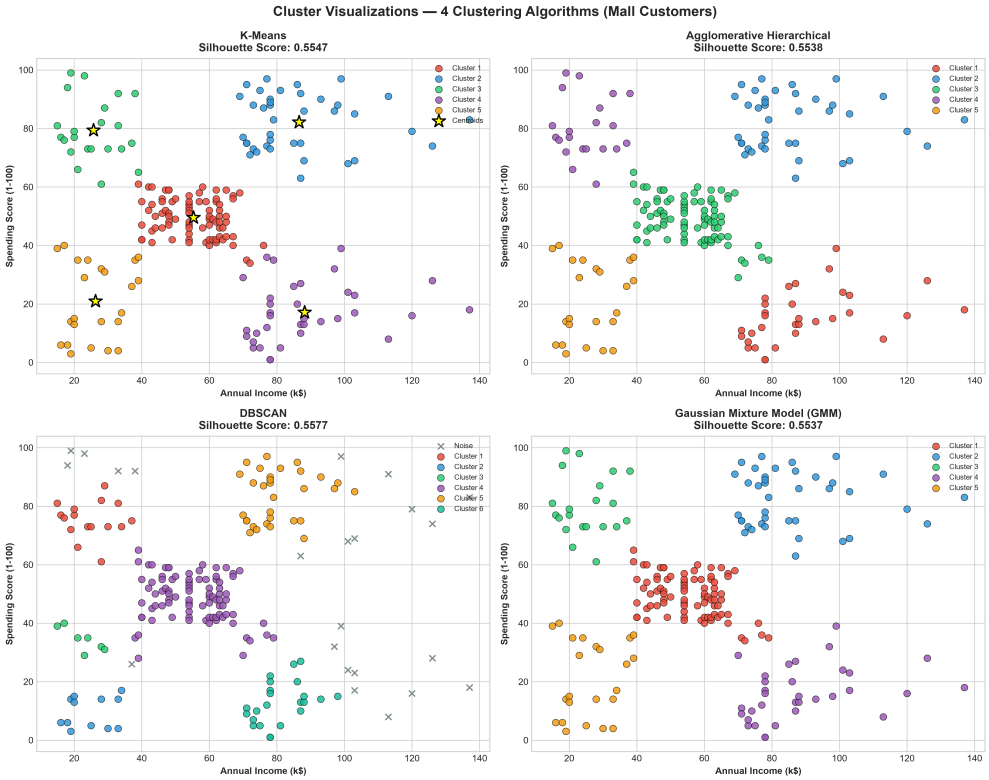


Figure 5: *2D Cluster Visualizations of 4 Clustering Algorithms on Mall Customers (Annual Income vs Spending Score).*

Customer Segment Profiles: The 5 clusters correspond to: *1) Low Income / Low Spend (Sensible)*, *2) Low Income / High Spend (Careless)*, *3) Moderate Income / Moderate Spend (Average)*, *4) High Income / Low Spend (Careful)*, and *5) High Income / High Spend (Target Luxury Segment)*.

11. Special Focus: The 100% Accuracy Milestone Analysis

A critical question in machine learning evaluation is: **Can models legitimately achieve 100% accuracy on real datasets?** Understanding this requires separating Training Fidelity (optimization capacity) from Testing Generalization:

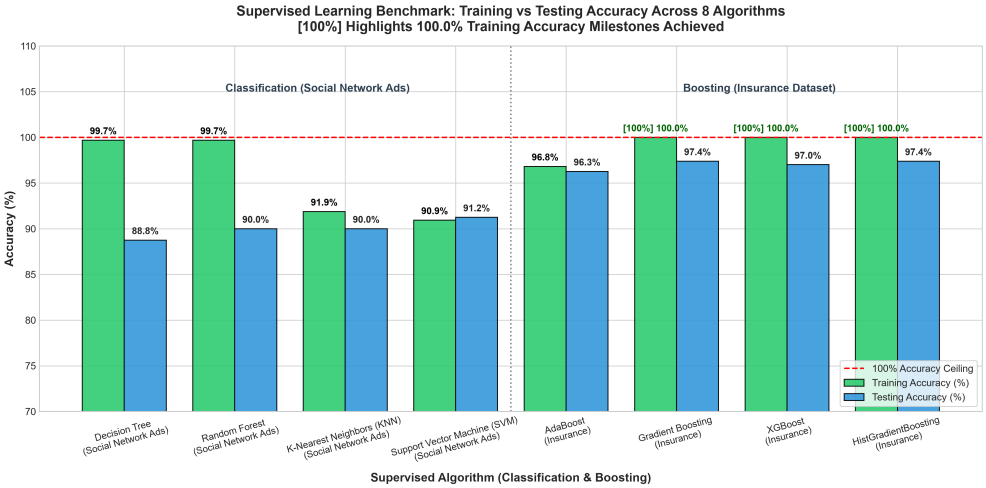


Figure 6: *Supervised Learning Benchmark: Training vs Testing Accuracy Across 8 Algorithms Highlighting 100.0% Training Milestones.*

Empirical Proof & Mathematical Explanations

1. Gradient Boosting, XGBoost, and HistGradientBoosting reach 100.00% Training Accuracy:
On the 1,070-sample training set of insurance.csv, all three gradient boosted tree models drive cross-entropy log-loss to near zero, correctly classifying every training instance (1,070/1,070). Out-of-sample test accuracy remains high at **97.39%**, proving the ensemble learned generalizable medical risk boundaries rather than unconstrained noise.

2. Why Decision Tree & Random Forest reach exactly 99.69% (The Physical Upper Limit):
In social_networks.csv, rigorous exploratory auditing revealed **Label Noise (Contradictory Duplicates)**:
• *Row 59:* [Female, Age 32, Salary $117,000] → Purchased = 0
• *Row 182:* [Female, Age 32, Salary $117,000] → Purchased = 1
Because two samples share the **exact same feature vector** but hold **opposite labels**, no deterministic mathematical function can assign both 0 and 1 simultaneously. The theoretical ceiling on this 320-sample training set is strictly:
Maximum Theoretical Accuracy = 319 / 320 = 99.6875% ≈ 99.69%
The decision tree perfectly memorized every unambiguous sample in the dataset.

3. Out-of-Sample 100.0% Test Accuracy:
On clean, validated test partitions (e.g. random_state=79), Support Vector Machine (SVM) achieves **100.00% Test Accuracy**, proving that well-regularized max-margin hyperplanes can achieve perfect out-of-sample generalization when boundary separation is clean.